

Reproduction of a Research Claim from Smith et al. (2016),
from The American Journal of Sociology

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Review Period: June 15 - June 22

View-only links to: [Original Paper](#), [Replication Analysis](#)

Note to Reviewers: Going forward, we will be implementing a new preregistration form. Please reference the [New Preregistration Form](#) and the [Reviewer Criteria Checklist](#) for instructions and specific criteria to help guide your review.

Privacy Statement: Other teams are making predictions about the outcomes of many different studies, not knowing which studies have been selected for replication. As a consequence, the success of this project requires full confidentiality of this peer review process. This includes privacy about which studies have been selected for replication and all aspects of the discussion about these replication designs.

Study Information

TITLE: Reproduction of a research claim from Smith et al, 2016, in American Journal of Sociology.

1. Authors (required)

AUTHORS:

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2. Description (optional)

DESCRIPTION:

The claim selected for replication from Smith et al (2016) is that classrooms with a larger share of natives show less ethnic homophily. This reflects the following statement from the abstract: "In order to effectively overcome ethnic segregation, we need to identify those factors within diverse settings that exacerbate the tendency toward ethnic homophily." The selected claim is tested using a two-step procedure. In a first step, the authors estimate ethnic homophily in classes using exponential random graph models (ERGMs) for each school class separately. In a second step, the authors examine the combined effects using a metaregression. The exact model we are asked to replicate comes from Table 3, Model 2. The focal estimate concerns the coefficient on the "Proportion natives" variable. The models predict the level of ethnic homophily in the classroom. Model 2 in Table 3 shows evidence in support of the hypothesis that ethnic homophily decreases with an increasing share of natives, $b = -.51$, $SE = .06$, $p < .001$.

3. Hypotheses (required)

- 3.1. *List specific, concise, and testable hypotheses. Please state if the hypotheses are directional or non-directional. If directional, state the direction. A predicted effect is also appropriate here. If a specific interaction or moderation is important to your research, you can list that as a separate hypothesis.*
- 3.2. **Example:** *If taste affects preference, then mean preference indices will be higher with higher concentrations of sugar.*
- 3.3. *The hypothesis for SCORE will be indicated as H^* . If you plan to do confirmatory research beyond the focal claim and test selected for the SCORE program, please make sure they are clearly delineated from the SCORE claim by numbering further hypotheses beginning with H_1 .*

HYPOTHESIS:

H^* (SCORE claim):

Classrooms with a larger share of natives show less ethnic homophily.

Design Plan

In this section, you will be asked to describe the overall design of your study. Remember that this research plan is designed to register a single study, so if you have multiple experimental designs, please complete a separate preregistration.

4. Study type (required; multiple choice, select one)

STUDY TYPE:

Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, natural experiments, and regression discontinuity designs.

5. Blinding (required; multiple choice, highlight applicable option(s))
- 5.1. *Blinding describes who is aware of the experimental manipulations within a study. Mark all that apply.*
- 5.1.1. *No blinding is involved in this study.*

BLINDING:

Does not apply.

6. Study design (required)
- 6.1. *Describe your study design. Examples include two-group, factorial, randomized block, and repeated measures. Is it a between (unpaired), within-subject (paired), or mixed design? Describe any counterbalancing required. Typical study designs for observation studies include cohort, cross sectional, and case-control studies.*
- 6.2. *Example: We have a between subjects design with 1 factor (sugar by mass) with 4 levels.*
- 6.3. *More info: This question has a variety of possible answers. The key is for a researcher to be as detailed as is necessary given the specifics of their design. Be careful to determine if every parameter has been specified in the description of the study design. There may be some overlap between this question and the following questions. That is OK, as long as sufficient detail is given in one of the areas to provide all of the requested information. For example, if the study design describes a complete factorial, 2 X 3 design and the treatments and levels are specified previously, you do not have to repeat that information.*

STUDY DESIGN:

Information about target population:

This is a cross-sectional analysis of classroom networks using the CILS4EU data. In the CILS4EU project, attribute and friendship data were collected for 18,716 adolescents around 14 years old in 958 school classes between October 2010 and June 2011. The schools were located in England, Germany, the Netherlands, and Sweden (Kalter et al. 2014). The authors of the original study used a two-step method in their original research, which we will also follow: we will first estimate ethnic homophily in classes using exponential random graph models (ERGMs) for each class separately. We will then examine the combined effects using a meta-regression.

7. Randomization (optional)
 - 7.1. *If you are doing a randomized study, how will you randomize, and at what level?*
 - 7.2. *Example: We will use block randomization, where each participant will be randomly assigned to one of the four equally sized, predetermined blocks. The random number list used to create these four blocks will be created using the web applications available at <http://random.org>.*
 - 7.3. *More info: Typical randomization techniques include: simple, block, stratified, and adaptive covariate randomization. If randomization is required for the study, the method should be specified here, not simply the source of random numbers.*

RANDOMIZATION:

The CILS4EU-Core team randomly selected schools and classrooms within a selected school. The procedure is documented in Kalter et al. (2014). No further randomization is used in our replication of the results.

Sampling Plan

In this section we'll ask you to describe how you plan to collect samples, as well as the number of samples you plan to collect and your rationale for this decision. Please keep in mind that the data described in this section should be the actual data used for analysis, so if you are using a subset of a larger dataset, please describe the subset that will actually be used in your study.

8. Existing data (required; multiple choice, highlight one option)

- 8.1.1. Registration prior to analysis of the data: As of the date of submission, the data exist and you have accessed it, though no analysis has been conducted related to the research plan (including calculation of summary statistics). A common situation for this scenario when a large dataset exists that is used for many different studies over time, or when a data set is randomly split into a sample for exploratory analyses, and the other section of data is reserved for later confirmatory data analysis.
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9. Explanation of existing data (optional)

- 9.1. *If you indicate that you will be using some data that already exist in this study, please describe the steps you have taken to assure that you are unaware of any patterns or summary statistics in the data. This may include an explanation of how access to the data has been limited, who has observed the data, or how you have avoided observing any analysis of the specific data you will use in your study.*

EXPLANATION OF EXISTING DATA:

The full dataset can only be accessed in the Secure Data Center of GESIS in Köln. We have now applied for access to the secure data center and have to schedule a visit later this spring. At the same time, a reduced version of the data can be obtained for offsite use after applying through GESIS. We have now access to the reduced version of the data, which however does not include information on country of origin. For now, we use this reduced version to examine missing values on the variables which are available (see section on Missing Data below), and also as a template to write our replication code.

Importantly, we do not and cannot conduct any replication analyses by running our code on this reduced version of the dataset, as the crucial information on country of origin is not included.

Please also note that the CILS dataset has been updated since the publication of the Smith et al. article. This update eliminated some coding errors, and also supplies a pre-prepared ISEI (International Socio-Economic Index of Occupational Status) variable. The ISEI variable was

constructed by hand in the original article. However, we will use the pre-prepared variable in our replication.

Code for the data preparation and analysis can be found here:

https://osf.io/nekdp/?view_only=d6003d6c12ac4970893a18099b8277a2

10. Data collection procedures (required)

DATA COLLECTION PROCEDURES:

Schools were selected with a probability proportional to their size, and the sample is stratified on the number of immigrant children attending the school so that the data include sufficient immigrant children. In the case that schools refused to participate, a similar school in terms of its school type or region was chosen within the same stratum as a replacement. In every school that participated, at least two classes of approximately 14-year-old students were selected using simple random sampling. The overall response rate among schools is 45% before replacement and 85% after replacement. These rates are comparable to other large-scale international school surveys such as the Program for International Student Assessment and the Trends in International Mathematics and Science Study. Moreover, as non-participating schools were replaced by schools similar in school type or region and in percentage of immigrant children, bias on a school level should not be a significant problem to our study's results and conclusions.

Parents were informed about the study in advance and could either opt in to the study (Germany) or opt out of the study (England, the Netherlands, and Sweden). To avoid lower participation rates in Germany, students were offered a €10 incentive in the case of participation. In addition to parental refusal, adolescents could be absent on the day of the interview or could decline to participate. The overall response rate among students was 85%

11. Sample size (required)

11.1. *Describe the sample size of your study. **How many units will be analyzed in the study?** This could be the number of people, birds, classrooms, plots, interactions, or countries included. If the units are not individuals, then describe the size requirements for each unit. If you are using a clustered or multilevel design, how many units are you collecting at each level of the analysis?*

SAMPLE SIZE:

The main analysis of the paper, and the hypothesis we are asked to replicate, concerns the meta-analysis of classrooms. CILS contains information on 958 classrooms. However, not all of these classrooms are included in the meta-analysis.

Instead, classrooms are excluded if they do not meet specific selection criteria, or if the classroom-level ERGM estimates do not converge.

These criteria are detailed in the section on “Exclusion Criteria” below.

For reference, after applying these criteria, the original paper uses 523 classrooms to test the hypothesis we are asked to replicate (see Model 2, Table 3, page 1247).

We note here that the ERGM package has been updated since the original article was published. The new ERGM algorithm may therefore yield different outputs, which could result in a different number of observations after applying the criteria for model convergence. Should this be the case, we will also rerun our analyses using the version of ERGM package used by the original studies’ authors (see “Deviations” section below).

12. Sample size rationale (optional)
 - 12.1. *This could include a power analysis or an arbitrary constraint such as time, money, or personnel.*
 - 12.2. **Example:** *We used the software program G*Power to conduct a power analysis. Our goal was to obtain .95 power to detect a medium effect size of .25 at the standard .05 alpha error probability.*
 - 12.3. **More information:** *This gives you an opportunity to specifically state how the sample size will be determined. A wide range of possible answers is acceptable; remember that transparency is more important than principled justifications. If you state any reason for a sample size upfront, it is better than stating no reason and leaving the reader to “fill in the blanks.” Acceptable rationales include: a power analysis, an arbitrary number of subjects, or a number based on time or monetary constraints.*

SAMPLE SIZE RATIONALE:

Does not apply.

13. Stopping rule (optional)

13.1. *If your data collection procedures do not give you full control over your exact sample size, specify how you will decide when to terminate your data collection.*

Secondary data analysis. Does not apply.

Variables

In this section you can describe all variables (both manipulated and measured variables) that will later be used in your confirmatory analysis plan. In your analysis plan, you will have the opportunity to describe how each variable will be used. If you have variables which you are measuring for exploratory analyses, you are not required to list them, though you are permitted to do so.

14. Manipulated variables (optional)

14.1. *Describe all variables you plan to manipulate and the levels or treatment arms of each variable. This is not applicable to any observational study.*

MANIPULATED VARIABLES:

Does not apply.

15. Measured variables (required)
 - 15.1. *Describe each variable that you will measure. This will include outcome measures, as well as any predictors or covariates that you will measure. You do not need to include any variables that you plan on collecting if they are not going to be included in the confirmatory analyses of this study.*
 - 15.2. **Example:** *The single outcome variable will be the perceived tastiness of the single brownie each participant will eat. We will measure this by asking participants 'How much did you enjoy eating the brownie' (on a scale of 1-7, 1 being 'not at all', 7 being 'a great deal') and 'How good did the brownie taste' (on a scale of 1-7, 1 being 'very bad', 7 being 'very good').*
 - 15.3. **More information:** *Observational studies and meta-analyses will include only measured variables. As with the previous questions, the answers here must be precise. For example, 'intelligence,' 'accuracy,' 'aggression,' and 'color' are too vague. Acceptable alternatives could be 'IQ as measured by Wechsler Adult Intelligence Scale' 'percent correct,' 'number of threat displays,' and 'percent reflectance at 400 nm.'*

MEASURED VARIABLES:

Our overall approach here is to attempt to construct variables as faithfully as possible going by what is described in the main text and appendices of the Smith et al. paper. Only in cases where the main text is vague or unclear will we refer to the original study's code. In practice, this means that if the main text and the original study's code were to diverge in some way, we would "default" to the main text.

Two sets of variables are used in the analysis. The first set of variables is used to estimate ethnic homophily measures at the classroom level. The second set of variables is used for the meta-analysis. We begin by describing the variables used in the first-stage (ERGM) estimation.

The ERGMs estimate whether a friendship tie exists between two students. Students received a class roster with the names and corresponding numbers of all their peers in class. From this roster, students choose their five best friends. All nominations are treated and modelled as directed ties. Therefore, a setup with $ij = 1$ and $ji = 0$ is treated as a friendship sent from i to j , but not as a friendship sent from j to i , preserving the structure of the friendship nomination data. In terms of the network adjacency matrix m , element $m[i, j] = 1$ and element $m[j, i] = 0$ in that case.

The primary variable of interest in the first stage relates to **same-ethnic background**. This is a binary measure of whether dyads (i.e., every potential pairing between students in the classroom) involve same-ethnic peers. Ethnicity is coded on the basis of the national origin of

students' parents. Native students are adolescents whose biological parents were both born in the host country. Immigrant adolescents are likewise coded to their parents' country of origin. When adolescents have mixed heritage, they are coded to the immigrant parent in the case of a mixed native-immigrant background. In case parents originate from different nonnative countries, students are coded to the mother's country (see p.1238 of the original paper).

The authors also include several controls in the ERGMs. First, a dummy variable is constructed indicating whether dyads involve students of the **same gender**.

A second control relates to the **popularity difference** between peers in a dyad. Here, the original paper is a bit vague on how this variable is constructed, writing only that "Adolescents were asked to nominate the most popular students in class, and we calculate the relative amount of incoming nominations an adolescent receives" (p.1240). Following the original study's code provided through SCORE, we have decided to code the popularity score as the total number of incoming nominations divided by the size of the network (network size is equivalent to the number of students who participated in the survey, but is not necessarily equivalent to the "real" size of the classroom since participation rate is not 100%. See section on "Missing Data" below). Finally the **popularity difference** is calculated as the difference between popularity scores for the students in each dyad.

The **socioeconomic difference** between peers in a dyad is calculated on the basis of the highest occupational status in an adolescent's household. The original paper constructed ISEI scores based on ISCO-08 data, but the newest update to CILS has pre-prepared the ISEI variable, and this is what we will use in our replication. Note that the original article somewhat lacks transparency about the fact that the CILS actually contains both a student and a parental questionnaire, and ISEI scores for both mother and father can be obtained from both questionnaires. We use the higher of the mother / father ISEI scores from the child questionnaire, if available. If this information is not available on the child questionnaire for both the mother or the father, we use the highest ISEI score as indicated in the parental questionnaire. If ISEI scores are missing for both parents on both questionnaires, we use the mean ISEI score in the classroom. As far as we can tell from the original study's code, this is the procedure followed by Smith et al. Once ISEI scores are assigned to students in this manner, **socioeconomic difference** is calculated as the absolute difference in scores between pairs in the dyad.

Beyond these individual-level variables, the ERGMs estimation uses several variables describing the network structure itself. The "construction" of these variables is handled automatically in the ERGMs package.

First, **edges** measures the total number of friendship ties and captures the baseline density of the network. Second, **mutual** reflects the tendency of adolescents to reciprocate friendships, and takes on a value of 1 if ego and alter both nominate each other as friends. The final variable is **GWESP** ("geometrically weighted edgewise shared partner"), which captures the effect of

transitivity, or the fact that ties between students are more likely to arise when they have more friends in common. In addition, GWESP Θ captures decreasing marginal returns of additional shared friends. In other words, the more friends you share, the more likely a friendship is, but the more friends you share, the smaller the effect of every additional friend on friendship becomes.

Note that all of these variables are determined by the structure of the *observed network*, which is a function of how many students fill out the survey. However, due to incomplete response rates, the *observed network* may be different from the *actual network* - that is, the “real” number of students in the classroom. We utilize the observed network because CILS does not have information on the actual network.

We note however that there is some leeway in constructing the observed network. For now, we simply flag the issue here. A more detailed discussion is provided in the section on “Missing Data” below.

We now turn to the second set of variables used in the meta-analysis. For our replication, the main variable of interest is the **proportion of natives** in the classroom. Ethnicity is defined in the same way as described for **same-ethnic background** above. There are additional controls for **class size**, and the “**mean socioeconomic status** (determined by the parental occupational status of all students in class)” (p.1241). Regarding **mean socioeconomic status**, it is unclear from the main text whether this variable utilizes all ISEI data (i.e., for both mothers and fathers in the household), or only data for the higher-SES parent (which is used to construct **socioeconomic difference** as described above). After consulting the original study’s code, we have determined that this variable is coded as a classroom-level average of the latter quantity.

Finally, the meta-analysis includes dummy variables for German, Dutch, and Swedish classes (with English classes as the reference category).

16. Indices (optional)

16.1. *If any measurements are going to be combined into an index (or even a mean), what measures will you use and how will they be combined? Include either a formula or a precise description of your method. If you are using a more complicated statistical method to combine measures (e.g. a factor analysis), you can note that here but describe the exact method in the analysis plan section.*

INDICES:

For clarity, all variables are described in the previous section. This section simply lists the variables which are indices or means.

The following variables are indices:

- Popularity score (used to calculate popularity difference)
- ISEI used to calculate socio-economic difference

The following variables are means:

- Proportion native
- Mean socioeconomic status

Analysis Plan

You may describe one or more confirmatory analysis in this preregistration. Please remember that all analyses specified below must be reported in the final article, and any additional analyses must be noted as exploratory or hypothesis generating.

A confirmatory analysis plan must state up front which variables are predictors (independent) and which are the outcomes (dependent), otherwise it is an exploratory analysis. You are allowed to describe any exploratory work here, but a clear confirmatory analysis is required.

17. Statistical models (required)

- 17.1. *What statistical model will you use to test each hypothesis? Please include the type of model (e.g. ANOVA, multiple regression, SEM, etc) and the specification of the model (this includes each variable that will be included as predictors, outcomes, or covariates). Please specify any interactions, subgroup analyses, pairwise or complex contrasts, or follow-up tests from omnibus tests. If you plan on using any positive controls, negative controls, or manipulation checks you may mention that here. Remember that any test not included here must be noted as an exploratory test in your final article.*
- 17.2. **Example:** *We will use a one-way between subjects ANOVA to analyze our results. The manipulated, categorical independent variable is 'sugar' whereas the dependent variable is our taste index.*
- 17.3. **More information:** *This is perhaps the most important and most complicated question within the preregistration. As with all of the other questions, the key is to provide a specific recipe for analyzing the collected data. Ask yourself: is enough detail provided to run the same analysis again with the information provided by the user? Be aware for instances where the statistical models appear specific, but actually leave openings for the precise test. See the following examples:*
- 17.3.1.1. *If someone specifies a 2x3 ANOVA with both factors within subjects, there is still flexibility with the various types of ANOVAs that could be run. Either a repeated measures ANOVA (RMANOVA) or a multivariate ANOVA (MANOVA) could be used for that design, which are two different tests.*
- 17.3.1.2. *If you are going to perform a sequential analysis and check after 50, 100, and 150 samples, you must also specify the p-values you'll test against at those three points.*

STATISTICAL MODELS:

The analysis proceeds in two steps. The first step uses ERGMs (with associated controls) to estimate the effect of **same-ethnic background** on friendship tie formation within classes. The same model is applied to all classrooms. Only those classrooms for which the ERGM converges are retained for the meta-analysis (see “Exclusion Criteria” below).

As noted above, the ERGMs package has been updated since the original article appeared. Our primary replication will use the new version of the ERGM package (see section on “Deviations” below). However, in case our results differ significantly from the original article (e.g., the new package could produce different homophily estimates and thereby affect which classrooms ultimately end up in the meta-analysis), we will also rerun all analyses using the original ERGMs package.

The second step examines the combined effects using a random-effects meta-regression. The meta-analysis models the tendency towards ethnic homophily (as captured by the classroom-level coefficients on **same-ethnic background**) as a function of **proportion of natives** (and associated controls).

According to the main text, the meta-analysis is conducted by weighting the coefficients on same-ethnic background by their variance (see p.1238). However this is incorrect: the inverse variance should be used. In fact, the authors themselves provide the correct description later in the same paragraph (“Classes in which homophily coefficients have larger standard errors contribute less to the average effect than classes with smaller standard errors.”) Note in addition that the original study’s code correctly implements the meta-analysis using the correct weights. Thus in this one instance where the main text and the original study’s code diverge, our replication will not follow what is written in the main text.

18. Transformations (optional)

18.1. *If you plan on transforming, centering, recoding the data, or will require a coding scheme for categorical variables, please describe that process.*

18.2. **Example:** *The “Effect of sugar on brownie tastiness” does not require any additional transformations. However, if it were using a regression analysis and each level of sweet had been categorically described (e.g. not sweet, somewhat sweet, sweet, and very sweet), ‘sweet’ could be dummy coded with ‘not sweet’ as the reference category.*

18.3. **More information:** *If any categorical predictors are included in a regression, indicate how those variables will be coded (e.g. dummy coding, summation coding, etc.) and what the reference category will be.*

TRANSFORMATIONS:

Does not apply

19. Inference criteria (optional)

- 19.1. *What criteria will you use to make inferences? Please describe the information you'll use (e.g. p-values, bayes factors, specific model fit indices), as well as cut-off criterion, where appropriate. Will you be using one or two tailed tests for each of your analyses? If you are comparing multiple conditions or testing multiple hypotheses, will you account for this?*
- 19.2. **Example:** *We will use the standard $p < .05$ criteria for determining if the ANOVA and the post hoc test suggest that the results are significantly different from those expected if the null hypothesis were correct. The post-hoc Tukey-Kramer test adjusts for multiple comparisons.*
- 19.3. **More information:** *P-values, confidence intervals, and effect sizes are standard means for making an inference, and any level is acceptable, though some criteria must be specified in this or previous fields. Bayesian analyses should specify a Bayes factor or a credible interval. If you are selecting models, then how will you determine the relative quality of each? In regards to multiple comparisons, this is a question with few "wrong" answers. In other words, transparency is more important than any specific method of controlling the false discovery rate or false error rate. One may state an intention to report all tests conducted or one may conduct a specific correction procedure; either strategy is acceptable.*

INFERENCE CRITERIA:

Criteria for a successful reproduction attempt for the SCORE project is a statistically significant effect ($\alpha = .05$, two tailed) in the same pattern as the original study. For this reproduction attempt, this criteria is met by a statistically significant negative effect of the share of natives on ethnic homophily in the meta-analysis regression model.

20. Data exclusion (optional)

20.1. *How will you determine what data or samples, if any, to exclude from your analyses? How will outliers be handled? Will you use any awareness check?*

20.2. **Example:** *No checks will be performed to determine eligibility for inclusion besides verification that each subject answered each of the three tastiness indices. Outliers will be included in the analysis.*

20.3. **More information:** *Any rule for excluding a particular set of data is acceptable. One may describe rules for excluding a participant or for identifying outlier data.*

DATA EXCLUSIONS:

The issue of data exclusions relates to the question of exactly which classrooms will be included in the second-stage meta-analysis. Exclusions will be made at two points in the analysis.

First, before any ERGMs are run, classrooms will be checked to see if they provide reliable sociometric data. According to p.1237 and footnote 10 on the same page, ERGMs will only be estimated for classrooms that:

- 1) Have at least 10 students
- 2) Have at least 2 natives and 2 non-natives
- 3) Have at least 2 males and 2 females
- 4) At least 75% of students participated in the network survey
- 5) No more than 10% of all nominations are invalid e.g., nominating peer 30 when there are only 15 students in class
- 6) No more than two students in class have never nominated anyone else in any of the 11 network-related items
- 7) No more than two students in class have never been nominated in any of the 11 network-related items

Note that these exclusion criteria are applied only with respect to the *observed network*. As mentioned above, there is some leeway in how the *observed network* can be constructed, which could impact which classrooms are excluded. Please see the section on “Missing Data” below.

It must also be noted that, although these are the exclusion criteria listed in the paper, our examination of the original study’s code provided through SCORE indicates that these criteria were implemented slightly differently in the original study. For example, the original study’s code imposed a threshold of 3 native and 3 non-native students, and 3 male and 3 female students, instead of 2. It also appears that no condition on the share of students participating in the network survey was implemented.

For our replication attempt, we will use the criteria listed above in 1-7, which are the criteria reported in the main text. In case our replication analysis does not reproduce the results from the original study, we will also run the analysis on the basis of the actual selection criteria implemented in the code of the original study.

Second, after ERGMs are run, classrooms will be excluded from the meta-analysis if the ERGMs do not converge or if the coefficients involved in the metaregressions show substantial problems. In particular, and in accordance with the strategy used in the original paper, we will exclude classrooms from the analysis in which a) the ethnic homophily coefficient is larger than 5 or smaller than -5, b) the standard error of this coefficient is larger than 5 or smaller than -5, or the change in the log-likelihood in the final phase of the ERGM estimation is larger than 5 (an indicator of model non-convergence). The original analyses do not conduct further assessments of model convergence.

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21. Missing data (optional)
 - 21.1. *How will you deal with incomplete or missing data?*
 - 21.2. **Example:** *If a subject does not complete any of the three indices of tastiness, that subject will not be included in the analysis.*
 - 21.3. **More information:** *Any relevant explanation is acceptable. As a final reminder, remember that the final analysis must follow the specified plan, and deviations must be either strongly justified or included as a separate, exploratory analysis.*

MISSING DATA:

The issue of missing data applies to the first stage ERGM estimation. A first issue with missingness arises from the fact that participation in CILS is not 100%. This could be due to absences, or parental / adolescent refusal to take the survey. This is an issue because we do not know the “real” number of students in a classroom (i.e., the size of the actual network). Thus, we construct our networks based on students who participated in the survey. We refer to this as the observed network.

An important consideration here is what to do about students who were nominated by others (thus, we know they “exist”), but who do not appear as respondents in the survey. Should they be included in the observed network? How these “missing” students are treated has consequences for modeling the structure of the network (e.g. GWESP), and also for applying the exclusion criteria. To illustrate, if a classroom has 8 students who answered the survey, but we know that there are (at least) 2 additional “missing” students, does this classroom meet the threshold of having class size ≥ 10 ?

It is not clear from the main text how this issue is handled. From the original study’s code, we can surmise that the observed network is constructed without these “missing” students. Thus, in our own code as well, students only enter the network if they have taken part in both the main CILS and sociometric questionnaire.

A secondary issue relates to missing values for students who otherwise completed the questionnaire. The table below lists the number of missing values for the student-level variables (i) gender, (ii) popularity score, (iii) highest household ISEI score. These are calculated using the reduced form of the dataset obtained through GESIS. Please note that, because we do not have access to the full dataset at this stage, we cannot determine the amount of missing data related to ethnicity.

Variable	Non-missing obs	Missing obs
Gender	11722	8
Popularity Score	11730	0
Highest HH ISEI	11123	607

In the version of the ERGM package used in the original paper, missing data results in list-wise deletion. It is presumably for this reason that the original paper substitutes missings in the highest parental ISEI with the classroom mean. The original paper also sets missing gender to female.

However, in the latest version of the ERGMs package, missing data can be handled without list-wise deletion. That said, in order to have analyses that are in line with the original paper, we also substitute missing values in the highest parental ISEI with classroom means just as the original paper does. Only for gender, we keep missing values in the analysis because the number of missing values is very low such that this divergence from the original analysis is unlikely to be consequential for final results.

22. Exploratory analysis (optional)

- 22.1. *If you plan to explore your data set to look for unexpected differences or relationships, you may describe those tests here. An exploratory test is any test where a prediction is not made up front, or there are multiple possible tests that you are going to use. A statistically significant finding in an exploratory test is a great way to form a new confirmatory hypothesis, which could be registered at a later time.*
- 22.2. **Example:** *We expect that certain demographic traits may be related to taste preferences. Therefore, we will look for relationships between demographic variables (age, gender, income, and marital status) and the primary outcome measures of taste preferences.*

EXPLORATORY ANALYSIS:

Not planned. If we find significant deviations from the original results, we will conduct further analyses to determine the source of these discrepancies.

Other

23. Deviations from the original study (required)
 - 23.1. Please describe any known deviations from the original study. Include discussion of whether and how these deviations might impact the results of this replication/reproduction attempt and its comparison to the original study.
24. Other (optional)
 - 24.1. If there is any additional information that you feel needs to be included in your preregistration, please enter it here. Literature cited, disclosures of any related work such as replications or work that uses the same data, or other context that will be helpful for future readers would be appropriate here.

DEVIATIONS AND OTHER INFORMATION:

The original study used the CILS4EU data of Wave 1 in the version 1.1.0 We will instead use version 1.2.0. The differences between the two versions are described in the CILS4EU-documentation which can be found here

<https://dbk.gesis.org/dbksearch/download.asp?db=E&id=59524>

Specifically, the coding of the socio-economic status of the parents from ISCO-08 to ESEI has been added in version 1.2.0. We will use this variable instead of the coding used by the original authors.

We will also use the updated ERGM package as a default for our analysis. In case of discrepancies in our results and the findings from Smith et al., we will run the code with the old ERGM package used to produce the output of Smith et al. 2016.
